SEMESTER 1 EXAMINATIONS 2022/2023

MODULE: CA119 - Data Science and Databases

PROGRAMME(S):

DS	BSc in Data Science
AP	BSc in Applied Physics
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 1,3,O

EXAMINER(S):

Mark Roantree	(Internal)	(Ext:5404)
Assoc. Prof. Sheila McBreen	(External)	External
Prof. Gerard O'Connor	(External)	External
Dr. Ziqi Zhang	(External)	External

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 (Entity-Relationship Modelling) [TOTAL MARKS: 25]

Climate scientists are creating a new experiment using air quality sensors in Dublin City and expect to generate large amounts of data. You are required to build their database but the first step will be to model their data using an Entity-Relationship model.
The different entities are the **Scientist** with names, different qualifications, their workplace (a university), and any other attributes you feel should be captured. The **Sensor** involved will generate data fields of time, temperature, and 3 different air quality readings PM2.5, PM5 and PM10 (which are just numbers). A scientist sets up a **Network** of these sensors at different locations around the city. A **Department** (representing a university department in which a scientist works) should be represented as a *weak* entity with attributes (FacultyName, Country, DateJoined) where DateJoined was the date they joined the project.

Q 1(a) [8 Marks]
Begin by creating the 4 entities shown in bold in the description above. 2 of the entities should have complex attributes; 1 entity should have a multi-valued attribute, and 1 entity should have a derived attribute.

Q 1(b) [8 Marks]
All scientists use all three sensor types in their experiments.
A network will have many scientists but scientists will only work on one network.
A scientist may or may not have a Department in which they work.
Now add all relationships to your E-R diagram.

Q 1(c) [6 Marks]

(i) What are the 2 types of Participation Constraints in Entity-Relationship modelling? Explain the difference between each type.

(ii) Provide an example of each type in your E-R diagram (If one does not exist, it is very easy to add one).

Q 1(d) [3 Marks]
Explain how the Department entity gets its key. Be clear about how the partial key is part of the overall key.

[End of Question 1]

QUESTION 2 (Relational Algebra) *[TOTAL MARKS: 25]*

Relation S

SID	FNAME	Course	Location	Year
201708	Alannah	DS2	Dublin	2021
201755	Beth	DS2	Dublin	2020
201812	Conor	DS2	Dublin	2019
201899	Dylan	CA4	Galway	2021
201905	Eric	CA4	Galway	2020
201910	Naomi	SS1	Cork	2019

Relation T

Institute	Location	Tutor
DCU	Dublin	Jennifer
UCC	Cork	Amy
UCG	Galway	Phil
Maynooth	Kildare	Una

Relation S represents prize winning students from different cities: SID is a primary key; Location is the county in which they live; Course is what they are studying.

Relation T represents the universities located in those cities and the Tutor responsible for managing the award.

Q 2(a) [6 Marks]

(i) Write a query in English which requires an INTERSECT and then express that query using the relational algebra.

(ii) Write the result into your answer sheet.

Q 2(b) [6 Marks]

(i) Write a query in English which requires an SET DIFFERENCE (MINUS) and then express that query using the relational algebra.

(ii) Write the result into your answer sheet.

Q 2(c) [6 Marks]

(i) Write the result of T × S to your answer book.

(ii) What is the degree of this new relation?

Q 2(d) [7 Marks]

(i) Write the result of T *left-outer-join* S to your answer book.

(ii) What is the degree of this new relation?

[End of Question 2]

QUESTION 3 (Normalisation) *[TOTAL MARKS: 25]*

DEPARTURES

FlightNo	Date	Destination	Pilots	Plane	Capacity	Load	Gate
FR666	24/10/2022	Birmingham	A. Murphy, B. Smith	B737	180	165	103
EI3228	24/10/2022	Glasgow	T.Trent, L. Byrne	A320	190	190	336
FR437	24/10/2022	Reus Salou	P. Henri, D. Gayet	B737	180	122	107
FR7156	24/10/2022	Madrid	G. Alonzo, G. Herrera	B737	180	178	104
EI638	24/10/2022	Brussels	D. Dunne, E. Evans	A320	190	188	422
LG4884	24/10/2022	Luxembourg	A. Claret, B. Tuna	B737	190	89	224
EI564	24/10/2022	Barcelona	G. Thompson, A. Lane	A321	210	210	418
EI528	24/10/2022	Paris CDG	N.Cave, J. Lydon	A320	190	160	414
FR288	24/10/2022	London STN	E Leclerc, E. Collins	B737	180	180	103
BA4473	24/10/2022	London LCY	R. Sunak, A.Mess	B737	180	175	204
TP1327	24/10/2022	Lisbon	D. Jota, P. Nunez	A320	190	180	304
EK164	24/10/2022	Dubai	A. Latif, B. Asad, C. Murphy	A330	250	242	410

The table above represents a sample of departures from Dublin airport. The primary key for this table is (FlightNo, Date) as each flight (FlightNo) departs only once per day. All plane types (Plane) have the same maximum seating (capacity) and all planes require 2 pilots except for the A330 which requires 3 pilots. Load represents the actual seat occupancy on the flight.

Q 3(a) [3 Marks]

- (i) Convert DEPARTURES to 1NF. Explain what you did and why.
- (ii) Show the new schema (only necessary to provide the schema structure).

Q 3(b) [5 Marks]

- (i) Test all tables for 2NF. How did you perform this test?
- (ii) Which table(s) was not in 2NF and why?

Q 3(c) [4 Marks]

Convert the schema to 2NF. Show the new schema (only necessary to provide the schema structure).

Q 3(d) [4 Marks]

- (i) How do you perform the test for 3NF?
- (ii) Does any part of the schema fail this test? Explain why.

Q 3(e) [9 Marks]

- (i) How do you fix a failed 3NF test so that the schema can conform to 3NF?
- (i) Does it require a change to this schema?
- (ii) Show the new schema <u>and include all data</u>.

[End of Question 3]

QUESTION 4 (Functional Dependencies) *[TOTAL MARKS: 25]*

Col1	Col2	Col3	Col4	Col5
127	abc	Green	A1	Yes
128	abd	Green	A2	Yes
129	abe	Green	A3	Yes
132	abe	Red	A7	No
133	abf	Green	B1	Yes
134	abf	Green	B3	Yes
140	abg	Red	B4	No
141	abh	Red	B5	Partial
148	abj	Red	B6	Partial
149	abk	Green	B9	Yes

You are required to locate functionality dependencies (FDs) where the only information you have is that **Col1** is the primary key.

In all questions below, |X| and |Y| means the cardinality of X and Y is representing FDs.

Q 4(a) [8 Marks]

(i) Describe the method for determining all rules X → Y where: |X| = 1 and |Y| = 1.
(ii) Does the FD Col1 → Col2 hold?
(iii) Does the FD Col2 → Col1 hold?
Explain your answers for (ii) and (iii)
(iv) Provide 2 FDs (not including any FD in (ii) or (iii) that hold. For this part of the question, |X| = 1 and |Y| = 1.

Q 4(b) [6 Marks]

Write the full list of potential FDs X → Y where:
|X| = 2 and X **must contain** Col4 *and*
|Y| = 1 *and*
X and Y are disjoint (cannot contain the same members).

Q 4(c) [5 Marks]

Test all potential FDs in part(b) to see if they hold. Explain why in each case.

Q 4(d) [6 Marks]

Assume rules where |X| can be any cardinality but CANNOT contain Col1 or Col4. Provide examples of 3 valid FDs. Explain why each FD holds.

[End of Question 4]

QUESTION 5 (SQL Query Expressions) *[TOTAL MARKS: 25]*

The relational schema shown below is part of a hospital database. The primary keys are underlined.

Patient (<u>patientNo</u>, patName, patAddr, DOB)
Ward (<u>wardNo</u>, wardName, wardType, noOfBeds)
Contains (<u>patientNo, wardNo, admissionDate</u>)
Drug (<u>drugNo</u>, drugName, costPerUnit)
Prescribed (<u>patientNo</u>, <u>drugNo</u>, unitsPerDay, <u>startDate</u>, finishDate)

5(a) [3 marks]
List all the patients admitted on 1st April 2022.

5(b) [4 marks]
Find the names of all the patients being prescribed 'Morphine'.

5(c) [3 marks]
List all the patients contained in the 'Surgical' ward.

5(d) [4 marks]
What is the maximum, minimum and average number of beds in a ward? Create appropriate column headings for the results table.

Q5(e) [6 marks]
For each ward that admitted more than 10 patients on April 1st 2022, list the ward number, ward type and number of beds in each ward.

Q5(f) [5 marks]
List the numbers and names of all patients and the drugNo and number of units of their medication. The list should also include the details of patients that are *not* prescribed any medication.

[End of Question 5]

[END OF EXAM]